# Physics 214/224, Practice Quiz 5, v2.0

Primary instructor: Yurii Maravin

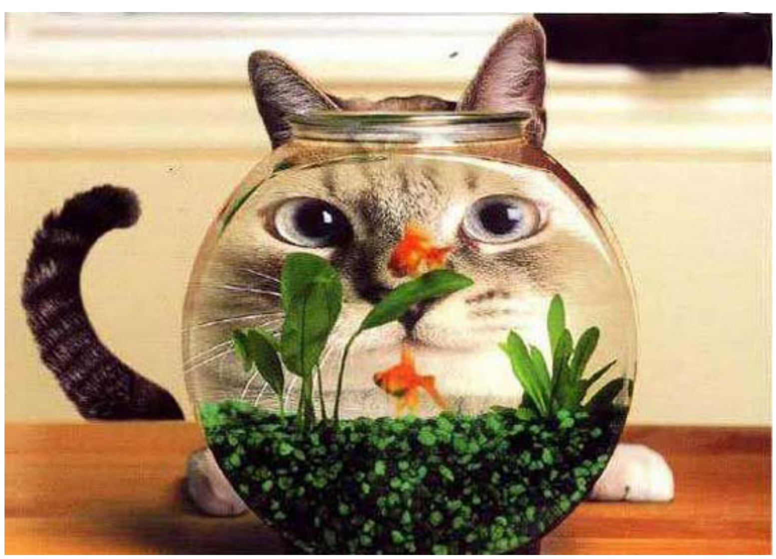
WID:______________________________ Name:________________________________________

**Instructions.** Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this academic work."

**For two quick points, circle your studio instructor:**

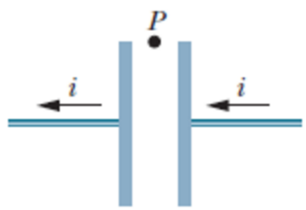
| | | |
|---|---|---|
| Lohman (TU 7:30) | Washburn (TU 9:30) | Maravin (TU 9:30) |
| Maravin (TU 11:30) | Lohman (TU 1:30) | Washburn (TU 3:30) |
| Lohman (WF 7:30) | Rodnova (WF 9:30) | Rodnova (WF 11:30) |

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.



## Part I: All questions to be answered on your Scantron card (48 points total)

The figure below shows a parallel-plate capacitor and the current in the connecting wire that is **discharging** the capacitor. In the next three questions you need to identify correct statements regarding the electric field and its flux between the capacitor's plates, direction of the displacement current, and the direction of magnetic field.

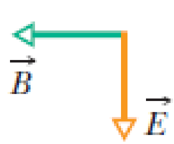



1. (3 points) Select the correct statement below:
   (a) The direction of the displacement current $i_d$ between the plates is leftward.
   (b) The direction of the displacement current $i_d$ between the plates is rightward.
   (c) The displacement current is zero, i.e., no direction.

2. (3 points) Select the correct statement below:
   (a) The magnetic field at point $P$ is directed into the page.
   (b) The magnetic field at point $P$ is directed out of the page.
   (c) There is no magnetic field at point $P$ because there is no actual current of charged particles between the plates.

3. (3 points) A current of 10 A is used to charge a parallel plate capacitor with square plates. If the area of each plate is 0.5 $m^2$, the displacement current through a 0.15 $m^2$ area wholly between the capacitor plates and parallel to them is
   (a) 1 A
   (b) 0.25 A
   (c) 3 A
   (d) 10 A
   (e) 5 A

4. (3 points) The figure below shows the electric and magnetic fields of an electromagnetic wave at a certain instant. What is the direction of the propagation of the wave?




   (a) to the left
   (b) to the right
   (c) into the page
   (d) out of the page
   (e) to the top

5. (3 points) Which of the following is NOT true for electromagnetic waves?
   (a) they transport energy
   (b) they transport momentum
   (c) they travel at different speeds in vacuum, depending on their frequency
   (d) they consist of changing electric and magnetic fields
   (e) they can be reflected

6. (3 points) We perceive intensity for sound wave as volume. What do we perceive intensity for light waves?
   (a) Color
   (b) Brightness
   (c) Opacity
   (d) Shininess

7. (3 points) Polarized light passes through a single polarizer with an intensity of 10.00 $mW/m^2$. The polarization plane of the light is aligned with the polarization axis of the polarizer. What is the intensity of the light upon exiting the polarizer?
   (a) 0.0 $mW/m^2$
   (b) 2.50 $mW/m^2$
   (c) 5.00 $mW/m^2$
   (d) 10.00 $mW/m^2$

8. (3 points) Which property of a light wave **does not** change as the wave passes from one transparent medium to another:
   (a) Frequency
   (b) Wavelength
   (c) Phase velocity
   (d) Amplitude

9. (3 points) Diverging lens can make
   (a) real images
   (b) virtual images
   (c) can do both real and virtual images

10. (3 points) Considering what you would observe on a nice day, which color of light do you think scatters most in the atmosphere?
    (a) Blue
    (b) Red
    (c) Yellow
    (d) White (all colors scatter the same)

11. (3 points) The physical phenomenon that makes fiber optic signal transmission possible is
    (a) Dispersion
    (b) Polarization by reflection
    (c) Small angle refraction
    (d) Total internal reflection

12. (3 points) You are looking at the mirror and notice that when you are close to it, you see yourself upright and magnified. As you move away from it, at one moment you see your image flip over and become inverted and de-magnified. This mirror is
    (a) convex
    (b) concave
    (c) planar
    (d) impossible... you are sleeping. Wake up!

13. (3 points) Interference of light is an evidence that
    (a) Light is electromagnetic in character
    (b) The speed of light is very large
    (c) Light does not obey conservation of energy
    (d) Light is a wave phenomenon
    (e) Light is a transverse wave

14. (3 points) Suppose one of two beams of green light ($\lambda = 510$ nm) split equally from the same laser travels 6.375 $\mu$m farther in reaching the same spot than the other. The intensity of light at that spot will be
    (a) Zero
    (b) The same as that of either beam by itself
    (c) Double that of either beam by itself
    (d) Four times that of either beam by itself

15. (3 points) What is the energy of a single photon of green light ($\lambda = 510$ nm)?
    (a) Zero
    (b) 2.4 eV
    (c) 5.1 eV
    (d) 1.96 mV
    (e) impossible to say

16. (3 points) If you shine a green light ($\lambda = 510$ nm) on a metal with a work function $\phi = 3.2$ eV, can you knock electrons from metal out?
    (a) yes, maximal kinetic energy of electrons would be about 2.1 eV
    (b) yes, maximal kinetic energy of electrons would be about 2.4 eV
    (c) yes, maximal kinetic energy of electrons would be about 0.3 eV
    (d) no, not possible

## Part II: Work out this part in the space provided. Show your work! (50 points total)

17. (14 points) A 1 cm tall object is placed 5 cm in front of a convex mirror with radius of curvature 20 cm.

    (a) (2 points) Find focal length of the mirror, including the sign

    (b) (2 points) Find location of the image

    (c) (2 points) Is image virtual or real? Is it inverted or erect?

    (d) (2 points) What is the height of the image?

    (e) (6 points) Check your answers with a ray diagram. Draw the mirror and the object. Use at least two rays to describe the location of the image. Please label rays or if you forgot the naming notation, briefly describe ray properties on the image.

18. (15 points) A 1 cm tall object is placed 5 cm in front of a symmetric biconcave lens constructed of $n = 1.55$ polystyrene plastic. Both surfaces of the lens have a radius of curvature $|R| = 10$ cm. Find the

   (a) (3 points) focal length of the lens, including the sign

   (b) (2 points) location of the image

   (c) (2 points) height of the image

   (d) (2 points) Is image virtual or real? Is image inverted or erect?

   (e) (6 points) Check your answers with a ray diagram.

19. (10 points) Like any good Kansan, you enjoy catching catfish with your bare hands (”noodling”). On one particular noodling expedition, you see a catfish in the water directly below you at a depth that exactly coincides with the location of your reflected image of your face. Your eyes are positioned 1.5 m above the surface of the water, which has an index of refraction $n = 1.33$.

   (a) (4 points) How far under the surface of the water is your reflected image?

   (b) (6 points) How far underneath the surface is the catfish?

20. (11 points) Does the spacing between fringes in a two-slit interference pattern increase, decrease, or stay the same under each of the following conditions? (You need to explain your answer for full credit!)

    (a) (2 points) If the slit separation is increased

    (b) (2 points) If the color of the light is switched from red to blue

    (c) (3 points) If the whole apparatus is submerged in generic beer (assume that index of refraction of generic beer is pretty much close to water, or $n \approx 1.33$)

    (d) (2 points) What happens to the interference pattern if you close one of the slits completely?

    (e) (2 points) What happens to the interference pattern if you shine blue light through one slit, and red through the other?

# Equations and Constants for Quiz 4

Note: you may detach this sheet from the quiz. You do not need to hand it in.

1. Point charge electric field
   (a) $\vec{E}_Q(\vec{r}) = \frac{kQ}{|\vec{r}-\vec{r}_Q|^2}\frac{\vec{r}-\vec{r}_Q}{|\vec{r}-\vec{r}_Q|}$;
   (b) $k = \frac{1}{4\pi\epsilon_0} = 9\times10^9$ N·m$^2$/C$^2$;
   (c) proton charge: $e = 1.6\times10^{-19}$ C.
2. Electric force: $\vec{F}_{1\to2} = Q_2\vec{E}_{Q_1}(\vec{r}_{Q_2})$;
   (a) Coulomb's Law: $\vec{F}_{1\to2} = \frac{kQ_1Q_2}{|\vec{r}_2-\vec{r}_1|^2}\frac{\vec{r}_2-\vec{r}_1}{|\vec{r}_2-\vec{r}_1|}$;
3. Superposition: $\vec{E}_{\text{TOT}}(\vec{r}) = \sum_n \frac{kQ_n}{|\vec{r}-\vec{r}_n|^2}\frac{\vec{r}-\vec{r}_n}{|\vec{r}-\vec{r}_n|}$ or $\int \frac{kdq(\vec{r}')}{|\vec{r}-\vec{r}'|^2}\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|}$.
4. Gauss's Law
   (a) $\oint \vec{E}\cdot d\vec{A} = \frac{1}{\epsilon_0}Q_{\text{enc}}$;
   (b) spheres: $4\pi r^2E = \frac{1}{\epsilon_0}\int_0^r 4\pi r'^2\rho(r')\,dr'$;
   (c) cylinders: $2\pi rLE = \frac{1}{\epsilon_0}\int_0^r 2\pi r'L\rho(r')\,dr'$;
   (d) conductors:
      i. $E = Q = 0$ inside.
      ii. On surface, $\vec{E}_\parallel = 0$; $\vec{E}_\perp = \sigma/\epsilon$.
5. Potential/Voltage and Potential Energy
   (a) $V_2 - V_1 = -\int_1^2 \vec{E}\cdot d\vec{r}$;
   (b) $V = \sum_n \frac{Q_n}{4\pi\epsilon_0|\vec{r}-\vec{r}_n|}$ or $\int \frac{dq(\vec{r}')}{4\pi\epsilon_0|\vec{r}-\vec{r}'|}$
   (c) $PE = U = qV$;
   (d) $V =$ constant on conductor.
6. Capacitance
   (a) $Q = CV$
   (b) $C_{\text{par.plate}} = \epsilon_0 A/d$
   (c) $C_{\text{dielectric}} = \kappa C_{\text{no dielectric}}$
   (d) $U = \frac{1}{2}CV^2 = Q^2/2C$
7. Current
   (a) $I = dQ/dt$
   (b) $\vec{J} = nq\vec{v}_{\text{drift}}$
   (c) $J = I/A$
   (d) $P = VI = V^2/R = I^2R$
8. Resistance
   (a) Ohm's law (micro) $\vec{J} = \sigma\vec{E}$
   (b) Ohm's law (macro) $V = IR$
   (c) $R = \rho L/A = L/\sigma A$
   (d) $P = dU/dt = VI = I^2R = V^2/R$
9. Circuits
   (a) $V_b - V_a = \sum_n \mathcal{E}_n - \sum_n i_n r_n$
   (b) $\sum_{in} I_n = \sum_{out} I_n$ (at any *junction*)
   (c) $\sum_n \mathcal{E}_n - \sum_n i_n r_n = 0$ (around any closed *loop*)
   (d)

| **component** | **series** | **parallel** |
|---|---|---|
| battery | $\mathcal{E}_1 + \mathcal{E}_2$ | $------$ |
| resistor | $R_1 + R_2$ | $(1/R_1 + 1/R_2)^{-1}$ |
| capacitor | $(1/C_1 + 1/C_2)^{-1}$ | $C_1 + C_2$ |

   (e) RC circuits:
      i. $\tau_{\text{RC}} = RC$.
      ii. $Q_{\text{charging}} = Q_{\max}\left(1 - e^{-t/\tau_{\text{RC}}}\right)$.
      iii. $Q_{\text{dis-charging}} = Q_{\max}e^{-t/\tau_{\text{RC}}}$.
10. Magnetic force/torque.
   (a) $\vec{F}_M = q\vec{v}\times\vec{B}$.
   (b) $\vec{v}\parallel\vec{B}\to$ no deflection.
   (c) $\vec{v}\perp\vec{B}\to$ circle with $R = \frac{mv}{qB}$.
   (d) $\vec{F} = I\vec{L}\times\vec{B}$.
   (e) $|\vec{\mu}| = IA$.
   (f) $\vec{\tau} = \vec{\mu}\times\vec{B}$.
   (g) $PE = -\vec{\mu}\cdot\vec{B}$
11. Magnetic fields
   (a) Biot-Savart law: $d\vec{B} = \frac{\mu_0 I}{4\pi}\frac{d\vec{\ell}\times(\vec{r}-\vec{s})}{|\vec{r}-\vec{s}|^3}$.
   (b) Ampere's Law: $\oint \vec{B}\cdot d\vec{s} = \mu_0 I_{\text{enc.}}$.
   (c) $\mu_0 = 4\pi\times10^{-7}$ T·m/A.
   (d) $B_{\text{wire}} = \mu_0 I/2\pi r$.
   (e) $B_{\text{solenoid}} = \mu_0 nI$.
   (f) Magnetism I: $\vec{B}_{TOT} = \vec{B}_{APP} + \vec{B}_{MAG}$.
   (g) Magnetism II: $\vec{B}_{TOT} = K_M\vec{B}_{APP} = (1+\chi_M)\vec{B}_{APP}$.
12. Faraday's Law
   (a) $\mathcal{E} = \oint \vec{E}\cdot d\vec{s} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \vec{B}\cdot d\vec{A}$;

(b) Lenz's law: induced current oppose flux changes.

13. Inductance

    (a) Self: $\Phi_B = LI$.
        i. solenoid: $L = \mu_0 N^2 A/\ell$.
    (b) Mutual: $\Phi_B^{(2)} = M_{12} I_1$.
        i. two solenoids: $M_{12} = \mu_0 N_1 N_2 A/\ell$.
    (c) Energy storage: $U = \frac{1}{2} L I^2$.
    (d) LR circuits:
        i. $I_{\text{energizing}} = I_{\max}\left(1 - \exp\left(-t/\tau_{RL}\right)\right)$.
        ii. $I_{\text{de-energizing}} = I_{\max} \exp\left(-t/\tau_{RL}\right)$.
        iii. $\tau_{RL} = L/R$.
    (e) Ideal transformer: $\mathcal{E}_2/\mathcal{E}_1 = I_1/I_2 = N_2/N_1$.
    (f) AC circuits
        i. $\mathcal{E}(t) = \mathcal{E}_0 \cos \omega t$.
        ii. $\mathcal{E}_{rms} = \sqrt{\langle \mathcal{E}^2(t) \rangle} = \mathcal{E}_0/\sqrt{2}$.
        iii. $\langle P \rangle = \mathcal{E}_{rms} I_{rms} \cos \phi$.
    (g) LC Oscillations
        i. $\omega = 2\pi f = \frac{1}{\sqrt{LC}}$
    (h) Series LCR circuit.
        i. $I(t) = \mathcal{E}_0 \cos(\omega t - \phi) / \sqrt{(X_L - X_C)^2 + R^2}$.
        ii. $X_L = 2\pi f L$.
        iii. $X_C = 1/(2\pi f C)$.
        iv. $\tan \phi = (X_L - X_C)/R$.
    (i) Damping (LCR circuit)
        i. $\omega_0 \to \omega' = \sqrt{\omega_0^2 - \frac{1}{4\tau^2}}$, $\tau = L/R$
        ii. if $1/(2\tau) > \omega_0$, system is "over-damped", no oscillations are possible
        iii. if $1/(2\tau) = \omega_0$, system is critically damped

14. Maxwell equations

    (a) Electric field flux $\Phi_E = \int_S \vec{E} \cdot d\vec{A}$
    (b) Displacement current $i_d = \epsilon_0 \frac{d\Phi_E}{dt}$
    (c) Gauss law for $\vec{E}$: $\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$
    (d) Gauss law for $\vec{B}$: $\oint_S \vec{B} \cdot d\vec{A} = 0$
    (e) Faraday law: $\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$
    (f) Maxwell-Ampere law: $\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$
    (g) $c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$

15. Wave properties

    (a) $A \cos(kx \mp \omega t)$ =wave moving to $\pm x$.
    (b) $\omega = 2\pi f, k = 2\pi/\lambda$.
    (c) $\lambda f = \lambda/T = \omega/k = v_P$.
    (d) Intensity: $I \sim |A|^2 = P_{in}/4\pi r^2$.
    (e) $\#dB = 10 \log_{10}(I/I_0)$.
    (f) Waves on a string:
        i. $\lambda_n = 2L/n$.
        ii. $v = \sqrt{T/\mu}$
    (g) Sound waves: $v = 1.3\sqrt{P/\rho}$.
    (h) EM waves:
        i. $\vec{E}(x,t) = \vec{E}_0 \cos(kx \mp \omega t)$.
        ii. $\vec{B}(x,t) = \vec{B}_0 \cos(kx \mp \omega t)$.
        iii. $v = c = E_0/B_0 = 1/\sqrt{\epsilon_0 \mu_0} = 3 \times 10^8$ m/s.
        iv. $\vec{E} \times \vec{B} \to$propagation direction.
        v. Polarization: $I_{\text{out}} = I_{\text{in}} \cos^2 \theta$.

16. Geometric optics

    (a) Reflection: $\theta_R = \theta_I$.
    (b) Refraction: $n_1 \sin \theta_1 = n_2 \sin \theta_2$.
    (c) Total internal reflection: $\sin \theta_C = 1/n$.
    (d) Refracting surface: $n_1/p + n_2/i = (n_2 - n_1)/R$.
    (e) Mirror: $1/p + 1/i = 1/f = 2/R$.
    (f) Lens: $1/p + 1/i = 1/f$.
    (g) Lens: $1/f = (n-1)(1/r_1 - 1/r_2)$.
    (h) Magnification: $m = -i/p$.

17. Interference effects

    (a) Constructive: $\Delta$(phase)= $2n\pi$; $\Delta$(path)= $n\lambda$.
    (b) Destructive: $\Delta$(phase)= $(2n+1)\pi$; $\Delta$(path)= $(n + 1/2)\lambda$.
    (c) 2-slit max: $\sin \theta = n\lambda/d$.
    (d) 2-slit min: $\sin \theta = (n + 1/2)\lambda/d$.
    (e) 1-slit min: $\sin \theta = m\lambda/a \, (m \neq 0)$.
    (f) Diffraction on circular opening: First minimum is at $\theta = 1.22\frac{\lambda}{D}$ with respect to center.
    (g) Uncertainty principle: $\Delta x \Delta \theta \simeq \lambda$.

18. Photoelectric effect

    (a) $\text{KE}_e = hf - \phi$

(b) $\lambda = 1234$ nm$/E$ (in eV)

19. velocity:
    (a) $\vec{v}_{AVG} = (\vec{r}_2 - \vec{r}_1)/(t_2 - t_1) \rightarrow \vec{v} = d\vec{r}/dt$ as $t_2 \rightarrow t_1$;
    (b) speed: $v = |\vec{v}|$.
20. acceleration:
    (a) $\vec{a}_{AVG} = (\vec{v}_2 - \vec{v}_1)/(t_2 - t_1) \rightarrow \vec{a} = d\vec{v}/dt$ as $t_2 \rightarrow t_1$;
    (b) constant acceleration:
        i. $\vec{v} = \vec{v}_0 + \vec{a}t$;
        ii. $\vec{r} = \vec{r}_0 + \vec{v}_0 t + \vec{a}t^2/2$.
    (c) Free fall:
        i. $\vec{a} = 0\hat{\imath} - g\hat{\jmath}$; $g = 9.8$ m/s$^2$.
21. Newton's Laws
    (a) $\vec{v}$ =constant $\Leftrightarrow \vec{F}_{NET} = 0$;
    (b) $m\vec{a} = \vec{F}_{NET}$;
    (c) $\vec{F}(1 \text{ on } 2) = -\vec{F}(2 \text{ on } 1)$.
22. Everyday forces
    (a) weight on Earth: $W = mg$;
    (b) tension: always pulls, constant throughout string;
    (c) normal force: $\perp$ to surfaces
    (d) friction: $\parallel$ to surfaces, opposes motion
        i. $f_K = \mu_K N$;
        ii. $0 \leq f_S \leq f_S^{MAX} = \mu_S N$;
        iii. $0 \leq \mu_K \leq \mu_S$;
    (e) spring: $F = -kx$;
23. Uniform curcular motion
    (a) $\vec{a}_C = v^2/r$ to center of circle;
    (b) period: $T = 2\pi r/v$;
24. Universal gravitation
    (a) $\vec{F}_{G,1\rightarrow 2} = \frac{-GM_1M_2}{|\vec{r}_2 - \vec{r}_1|^2}\frac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|}$; $G = 6.7 \times 10^{-11}$ N$\cdot$kg$^{-2}\cdot$m$^2$;
    (b) Kepler's 3rd law: $T^2 = \frac{4\pi^2}{GM}r^3$.
25. Work and Energy
    (a) $W = \int \vec{F} \cdot d\vec{r} \rightarrow \int F\cos\phi dx\,(1D) \rightarrow Fd\cos\phi\,(F \text{ constant})$;
    (b) $KE = \frac{1}{2}mv^2$;
    (c) $W = \Delta(KE) = KE_2 - KE_1$ (all forces);
    (d) $PE = -W$ (conservative forces);
    (e) $PE = mgy$ (gravity near earth surface);
    (f) $PE = -Gm_1m_2/r$ (all gravity);
    (g) $PE = \frac{1}{2}kx^2$ (spring);
    (h) $KE + PE$ =constant (conservative forces);
26. Momentum
    (a) $\vec{P} = \sum m_n \vec{v}_n$;
    (b) $\frac{d\vec{P}}{dt} = \vec{F}_{EXT}$;
    (c) impulse: $\vec{J} = \Delta\vec{P} = \vec{F}_{EXT}\Delta t$;
    (d) $\vec{F}_{EXT} = 0 \rightarrow \vec{P}$ =constant;
27. center of mass
    (a) few objects: $\vec{R}_{CM} = \sum m_n \vec{r}_n/M$;or $\int \vec{r}dm/M$;
    (b) $\vec{P} = M\frac{d\vec{R}_{CM}}{dt} = M\vec{V}_{CM}$;
    (c) $\vec{F}_{EXT} = 0$ and $\vec{P} = 0 \rightarrow \vec{R}_{CM}$ =constant.
28. Two-body collisions.
    (a) elastic: conserves $KE$;
    (b) inelastic: not elastic;
29. Rotation equations
    (a) For cloning translational formulas:
        i. $x \rightarrow \theta; \vec{v} \rightarrow \vec{\omega} = \frac{d\vec{\theta}}{dt}$;
        ii. $\vec{a} \rightarrow \vec{\alpha} = \frac{d\vec{\omega}}{dt}; m \rightarrow I = \sum m_n r_n^2$;
        iii. $\vec{F} \rightarrow \vec{\tau} = \vec{r} \times \vec{F}$;
        iv. $\vec{p} \rightarrow \vec{L} = \vec{r} \times \vec{p} = I\vec{\omega}$;
    (b) $KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$;
    (c) rolling without slipping:
        i. $v = \omega r$;
        ii. $a = \alpha r$.
    (d) angular momentum conservation
        i. $\frac{d\vec{L}}{dt} = \vec{\tau} = I\vec{\alpha}$;
        ii. $\vec{\tau} = 0 \rightarrow \vec{L}$ = constant;
        iii. $\vec{\tau} = 0 \rightarrow \sum I_n \vec{\omega}_n$ = constant.
30. Static equilibrium
    (a) $\sum \vec{F}_n = 0$;
    (b) $\sum \vec{\tau}_n = 0$.

31. Equilibrium and PE

    (a) $F(x) = -d(PE)/dx$;
    (b) $d^2(PE)/dx^2 > 0 \rightarrow$stable.

32. Fluids

    (a) $\rho = M/V$;
    (b) $P = F/A$;
    (c) $P = P_0 + \rho_F gd$;
    (d) $F_{\text{buoy}} = \rho_F g V_{\text{disp}}$(Archimedes);
    (e) $\nu A =$ constant (Venturi);
    (f) $\frac{1}{2}\rho v^2 + P + \rho g y =$ constant (Bernoulli).

33. Thermodynamics

    (a) Ideal gas: $PV = nRT$;
    (b) Ideal gas: $\langle KE \rangle = 3kT/2$.
    (c) Ideal gas: $v_{RMS} = \sqrt{3kT/m}$.
    (d) $\Delta L/L = \alpha \Delta T$ (thermal expansion);
    (e) $\Delta Q = Mc\Delta T$ (specific heat);
    (f) $\Delta Q = ML$ (latent heat, to change phase).
    (g) $\frac{dQ}{dt} = \frac{\kappa A(T_{HI} - T_{LO})}{L}$. (heat conduction)
    (h) $\frac{dQ}{dt} = \sigma \epsilon A \left(T_{HI}^4 - T_{LO}^4\right)$.(thermal radiation)
    (i) $0^{th}$ Law: eventually, $T_1 \rightleftharpoons T_2$;
    (j) $1^{st}$ Law: $\Delta U = \Delta Q - \Delta W$.
    (k) $2^{nd}$ Law (mechanic's version): $\epsilon = \frac{W_{OUT}}{Q_{IN}} \leq 1 - \frac{T_{LO}}{T_{HI}}$(heat engine).
    (l) $2^{nd}$ Law (entropy version): $\Delta S = \frac{\Delta Q}{T} \geq 0$ (isolated system).
    (m) Random walk: $d_{RMS} \propto \sqrt{t}$.